

OPERATIONAL GEOLOGY CHART
South IRAQ Stratigraphic Column

SYSTEM	SERIES (GROUP)	STAGE	FORMATION	MEMBER	GENERALIZED LITHOLOGY	Lithologies Types For Each Formation, Marker (Pick Criteria) For The Formations Tops, Lithologies Description, Average TVD Thickness For Each Formation, Estimated Formation Pore Pressure, Recommendation for Drilling Mud Weight, Down Hole Drilling Problems, Oil/Gas Reservoir Zones And Recommendation for Casing Points.
TERTIARY	PLIOCENE LATE MIOCENE		DIBDIBA			Dibdiba Formation : It's usually exposed at surface, It is composed mainly of Sand / Gravel intercalation with Minor Streaks of Limestone (Limestone & Claystone). Average TVD Thickness of Dibdiba Formation : +/- 225 m. (Heaving / Hole collapse / Caving) is highly possible which will affect hole stability & casing-cement quality in Dibdiba Formation. Sand / Gravel : Lse qtz gr, clss, transl - transp, m - crs gr, occ/ v.crs gm, sub ang - sub rd, p - mod strid, calc cmt lip, no vis por under visual exam. Limestone: Wh - off wh, sft - mod hd, crypto xln - fn xln, foss lip, wash lip, no vis por under visual exam. Marl: Lt gy, dk gy, med gy, grnsh gy, sft - mod hd, crypto xln, atky, hi calc occ/ grgd to hi arg limestone, no vis por under visual exam. Claystone: Rdsh brn, grnsh gy, sft, bkly - sbkly, stky, sil calc, washable lip. Gypsum: Shale with trace - transp, gray, mod hd, above shale. Estimated Fm. PP: 64 - 65 pcf (1.03 - 1.04 gm/cc) EMW Recommended drilling MW: 65 - 67 pcf (1.04 - 1.07 gm/cc)
						Lower Fars Formation : It is composed mainly of Marl and/or Limestone intercalation with Minor streaks of (Sandstone, Anhydrite & Claystone). It is picked by appearance of Thick layer of Marl and/or Limestone with Minor streaks of Sandstone. Average TVD Thickness of Lower Fars Formation : +/- 120 m. Caving is highly possible & also issues with plastic movement could be occurred (Marl & clay) in (Lower Fars Formation). Marl: Lt gry, occ/ grnsh gy, sft - mod hd, crypto xln, stky, wash lip, sdy lip, hi calc occ/ grgd to hi arg Limestone, no vis por under visual exam. Limestone : Lt gry - off wh, lt brn, mod hd, crypto xln - fn xln, foss lip, pyrc lip, sdy lip, hi arg, vuggy lip, no vis por under visual exam. Sandstone : Lse qtz gr, clse, transl - transp, occ/ med - crs grn, sub ang - sub rd, p - mod strid, calc cmt lip, foss lip, no vis por under visual exam. Anhydrite : Miky wh - wh, transp lip, smky wh lip, mod - hd - occ/ sft, psd, amorph. Claystone : Gy - grnsh gy, rdsh brn, sft, mod hrd lip, bkly - sbkly, stly lip, sil calc, stky, washable lip. Note : Lower Fars Fm. contains heavy oil in Nahr Ibn Umer field. Estimated Fm. PP: 64 - 65 pcf (1.03 - 1.04 gm/cc) EMW Recommended drilling MW: 65 - 67 pcf (1.04 - 1.07 gm/cc)
	MIDDLE MIOCENE	SERRAVALLIAN AQUITAN	LOWER FARIS			Ghar Formation : It is composed mainly of Sand / Gravel & Sandstone interbedded with sandy Limestone. It is picked by appearance of Sand / Gravel & Sandstone with Fast ROP. Average TVD Thickness of Ghar Formation : +/- 155 m. Sand / Gravel : Lse qtz gr, clse, transl - transp, m - crs gr, occ/ v.crs grn, sub ang - sub rd, p - mod strid, calc cmt lip, foss lip, no vis por under visual exam. Sandstone : Lse qtz gr, clse, transl - transp, occ/ yes, fribl - mod hd, vln xln - vln gr, mod - hi arg, sbang - sbnd, p - mod strid, calc cmt lip, foss lip, no vis por under visual exam. Limestone : Lt gry, off wh, lt brn, mod hd, crypto xln - fn xln, foss lip, sdy lip, no vis por under visual exam. Estimated Fm. PP: 64 - 65 pcf (1.03 - 1.04 gm/cc) EMW Recommended drilling MW: 64 - 65 pcf (1.03 - 1.04 gm/cc)
						Dammam Formation : It is composed mainly of Dolomite and/or Limestone occasionally intercalation with very minor streaks of (Anhydrite, Claystone & Marl). It is picked by appearance of carbonate (Dolomite or Limestone) with Slow ROP. Average TVD Thickness of Dammam Formation : +/- 240 m. partial / Complete Mud losses of circulation is highly possible while drilling Dammam Formation. Dolomite : Crmpy wh, off wh, tan, lt brn, mod hd - hd, microx - v.fxn, vugy lip, occ/ foss, suc tex, no vis por under visual exam. Limestone : Tansh wh - wh, occ/ tansh brn, mod hd - hd, mic xln - v.fxn, sil arg, vuggy lip, occ/ foss, dolc no vis por under visual exam. Anhydrite : Miky wh - wh, off wh, mod - hd - hd, psd, amorph, v Gypsum. Claystone : Tan - tansh wh, crmy, occ/ lt brn, mod hd - hd, micro xln - crypto xln, sil arg lip, occ/ sil arg, rr pyrc. Marl : Gnah gy, sft - mod hd, crypto xln - fn xln, foss lip, no vis por under visual exam. Limestone : Lt gry - med gy, grnsh gy, rr rdsh brn, sft, mod hrd lip, bkly - sbkly, stly lip, sil calc, stky, washable lip. Note : Dammam Fm. Could be contains Fractured (Dolomite and/or Limestone) which is leads Complete lost of circulation. Dammam Formation Casing Point : It is a regular casing point at south IRAQ, It is highly recommended to set the casing +/- 15 m into Dammam Formation top to prevent shallower formations collapsing. Estimated Fm. PP: 64 - 65 pcf (1.03 - 1.04 gm/cc) EMW Recommended drilling MW: 65 - 67 pcf (1.04 - 1.07 gm/cc)
	MIDDLE EOCENE	PRIABONIAN LUTETIAN	DAMMAM			Rus Formation : It is composed mainly of Anhydrite intercalation with very minor streaks of (Dolomite and/or Limestone). It is picked by appearance of Anhydrit Slow ROP. Average TVD Thickness of Rus Formation : +/- 150 m. Anhydrite : Miky wh - wh, smky wh lip, md hd - hd, occ/ sft, psd, amorph. Dolomite : Tan, crmy, tansh wh, occ/ lt brn, mod hd - hd, mic xln - v.fxn xln, vuggy, sil arg, no vis por under visual exam. Limestone : Lt gry - dk brnsh gy, dk brn, mod hd - hd, micro xln - v.fxn xln, sil arg, pyrc lip, no vis por under visual exam. Estimated Fm. PP: 64 - 65 pcf (1.03 - 1.04 gm/cc) EMW Recommended drilling MW: 65 - 68 pcf (1.04 - 1.11 gm/cc)
		YPRESIAN	RUS			Umm Er Radhuma Formation : It is composed mainly of Carbonates (Dolomite and/or Dolomitic Limestone and/or Limestone) interbedded with thin streaks of Carbonates. It is picked by appearance of Carbonates (Dolomite and/or Dolomitic Limestone and/or Limestone) with Fast ROP. Average TVD Thickness of Umm Er Radhuma Formation : +/- 450 m. Sulphurous water flow (and / or) partial / Complete Mud losses of circulation is highly possible in Umm Er Radhuma Formation. Dolomite : Tan - tansh wh, crmy, occ/ lt brn, mod hd - hd, microxln - v.fxn, vug lip, lip, occ/ sil arg, rr pyrc. no vis por under visual exam. Limestone : Lt gry - off wh, lt brn, mod hd - hd, micro xln - crypto xln, sil arg lip, no vis por under visual exam. Dolomitic Limestone : wh - off wh, occ/ gsh brn, microxln - v.fxn, mod hd - hd, sil arg, no vis por under visual exam. Anhydrite : Miky wh - wh, smky wh lip, md hd - hd, psd, amorph. Shale : Gy, grnsh gy, lt gy, frm - mod frm, occ/ sft, bkly - sb bkly, stky, dolc, occ/ stly. Note : Umm Er Radhuma formation containing water aquifer. Estimated Fm. PP: 64 - 65 pcf (1.03 - 1.04 gm/cc) EMW Recommended drilling MW: 68 - 71 pcf (1.09 - 1.14 gm/cc)
	PALEOCENE - EARLY EOCENE	THANETIAN DANIAN	UMM ER RADHUMA			Tayarat Formation : It is composed mainly of Dolomite interbedded with (Limestone and Anhydrite) layers. It is picked by appearance of a thin Shale layer at top of Tayarat Fm. with Slow ROP. Average TVD Thickness of Tayarat Formation : +/- 260 m. Sulphurous water flow (and / or) partial / Complete Mud losses of circulation is possible in Tayarat Formation. Shale : Dk gy, med gy, grnsh gy, frm - mod frm, occ/ sft, bkly - sb bkly, hi calc occ/ grgd to Marl. Dolomite : Lt gry - dk brnsh gy, dk brn, occ/ off wh, lt tan, m hd - hd, micro xln - v.fxn xln, suc tex, locally abhydritc, vugy, hi arg, no vis por under visual exam. Limestone : Tan, tansh brn, tansh gy, occ/ off wh, m hd - hd, micro xln - v.fxn xln, occ/ suc tex, vugy, dolc, hi arg grgd to Marl, no vis por, no vis por under visual exam. Anhydrite : Miky wh - wh, smky wh lip, md hd - hd, psd, amorph. Estimated Fm. PP: 64 - 65 pcf (1.03 - 1.04 gm/cc) EMW Recommended drilling MW: 68 - 71 pcf (1.09 - 1.14 gm/cc)
		MAASTRICHTIAN	TAYARAT			Shiraniash Formation : It is composed mainly of Limestone interbedded with thin layers of Marl. It is picked by appearance of Argillaceous Limestone grading to Marl with Slow ROP. Average TVD Thickness of Shiraniash Formation : +/- 105 m. Bit balling, Sticking & Tight hole may occur while drilling Shiraniash Formation. Limestone : Lt gy, off wh, tansh brn, crmy wh, med gy, mod hd, occ/sft, micro xln - crypto xln, dolc, gluc lip, hi arg grgd to Marl, no vis por under visual exam. Marl : Lt gy, gshy blk - dk gy, sft, occ/ mod hd, crypto xln, stky, wash lip, no vis por under visual exam. Estimated Fm. PP: 64 - 65 pcf (1.03 - 1.04 gm/cc) EMW Recommended drilling MW: 69 - 71 pcf (1.11 - 1.14 gm/cc)
	CRETACEOUS	UPPER	CAMPAIAN			Hartha Formation : It is composed mainly of Limestone interbedded with Dolomite & occasionally intercalation with streaks of Shale. It is picked by appearance of Chalky Limestone and/or Dolomite with Slow ROP. Average TVD Thickness of Hartha Formation : +/- 195 m. partial / Complete Mud losses of circulation is highly possible in Hartha Formation. Dolomite : Tansh brn, lt tan, lt gy, mod hd, v.fxn - micro xln, suc tex lip, vugy, sil arg, pyrc lip, no vis por under visual exam. Limestone : Lt gry - off wh, wh, chky, occ/ lt brn, mod hd - hd, crypto xln - v.fxn xln, sil dolc, sil arg lip, gluc lip, occ/ pyrc, no vis por under visual exam. Shale : Med gy, grnsh gy, lt grn, frm - mod frm, occ/ sft, bkly - sb bkly, occ/ sb flky, sil calc, stly lip. Estimated Fm. PP: 64 - 65 pcf (1.03 - 1.04 gm/cc) EMW Recommended drilling MW: 70 - 73 pcf (1.12 - 1.17 gm/cc)
						Sadi Formation : It is composed mainly of Limestone. It is picked by appearance of Chalky Argillaceous Limestone with Fast ROP. Average TVD Thickness of Sadi Formation : +/- 300 m. Tight hole & Sticking may occur while drilling Sadi Formation. Limestone : Off wh - wh, crmy wh, chky, lt gy, occ/ lt brn, mod hd - hd, crypto xln - v.fxn xln, sil dolc, foss lip, gluc lip, occ/ vuggy, sil - mod arg, no vis por under visual exam. Sadi Formation Casing Point : It is a regular casing point at south IRAQ, It is recommended to set the casing +/- 15 m into Sadi Formation top to seal off the Upper formations which has Mud Losses. Note : Sadi formation is an oil reservoir in some fields. Estimated Fm. PP: 64 - 65 pcf (1.03 - 1.04 gm/cc) EMW Recommended drilling MW: 70 - 73 pcf (1.12 - 1.17 gm/cc)
						Tanuma Formation : It is composed mainly of Shale occasionally intercalation with very minor streaks of Limestone. It is picked by appearance of Shale with Fast ROP. Average TVD Thickness of Tanuma Formation : +/- 50 m. Caving Shale & Tight hole may occur while drilling Tanuma Formation. Shale : Lt gy, dk gy - dk brnsh gy, grnsh gy - grn, occ/ blk, frm - mod frm, fssa - sb fssa, occ/ sb flky, stky, wash, pyrc lip, hi calc grgd to Marl and/or hi arg Lst. Limestone : Lt gy - off wh, wh, brnsh gy, occ/ lt brn, mod hd - hd, crypto xln - v.fxn xln, sil dolc, occ/ pyrc, occ/ hi arg grgd to hi calc Shale, no vis por under visual exam. Estimated Fm. PP: 64 - 65 pcf (1.03 - 1.04 gm/cc) EMW Recommended drilling MW: 71 - 73 pcf (1.14 - 1.17 gm/cc)
						Khasib Formation : It is composed mainly of Argillaceous Chalky Limestone intercalation with very thin beds of Shale . It is picked by appearance of Argillaceous Chalky Limestone with Slow ROP. Average TVD Thickness of Khasib Formation : +/- 45 m. Limestone : Lt gy - med gy, occ/ off wh, chky, mod hd - hd, micro xln - crypto xln, hi arg occ/ grgd to Marl, occ/ pyrc, no vis por under visual exam. Shale : Dk gy, grnsh gy, frm - mod frm, fssa - sb fssa, occ/ sb flky, stky, sil calc. Note : Khasib formation is an oil reservoir in some fields. Estimated Fm. PP: 64 - 65 pcf (1.03 - 1.04 gm/cc) EMW Recommended drilling MW: 72 - 73 pcf (1.15 - 1.17 gm/cc)
		MIDDLE	SANTONIAN CONCIAN			Mishrif Formation : It is composed mainly of Limestone occasionally with very Thin stringers of Shale. It is picked by appearance of Limestone with Slow ROP. Average TVD Thickness of Mishrif Formation : +/- 143 m. Limestone : Off wh - wh, crmy wh, sft - mod hd, microxln - v.fxn xln, gluc, vugy lip, sil dolc, no vis por under visual exam. Shale : Grnsh gy, dk gy, frm - mod frm, fssa - sb fssa, occ/ sb flky, stky, hi calc. Note : Mishrif Fm. is an oil reservoir in some fields. Estimated Fm. PP: 64 - 65 pcf (1.03 - 1.04 gm/cc) EMW Recommended drilling MW: 72 - 73 pcf (1.15 - 1.17 gm/cc)
						Rumaila Formation : It is composed mainly of Argillaceous Chalky Limestone occasionally with very minor streaks of Shale. It is picked by appearance of Argillaceous Chalky Limestone or Highly calcareous Shale with Slow ROP. Average TVD Thickness of Rumaila Formation : +/- 90 m or +/- 140 m. Limestone : Off wh, lt gy, tan gy, buff, chky, sft - mod hd, cryptoxln - microxln, pyrc lip, hi arg, occ/ grgd to marl, no vis por under visual exam. Shale : Grnsh gy, med - dk gy, frm - mod frm, bkly - sb bkly, stky, wash, hi calc occ/ grgd to hi arg Lst. Note : Rumaila formation is an oil reservoir in some fields. Estimated Fm. PP: 64 - 66 pcf (1.03 - 1.06 gm/cc) EMW Recommended drilling MW: 72 - 74 pcf (1.15 - 1.19 gm/cc)
						Ahmadi Formation : It is composed mainly of Shale & Limestone interbeds. It is picked by appearance of Shale followed by Argillaceous Limestone with Fast ROP. Average TVD Thickness of Ahmadi Formation : +/- 140 m. Caving Shale may occur while drilling Ahmadi Formation. Ahmadi Formation is divided into 2 parts (upper part is composed mainly of Shale with Minor Streaks of Limestone & the lower part is composed mainly of Limestone with Minor Streaks of Shale). Limestone : Tan - tansh wh, lt gy, occ/ brn, sft - md hd, microxln - crypto xln - v.fxn xln, suc tex lip, occ/ sil arg, no vis por under visual exam. Shale : Brnsh gy, grnsh gry, occ/ lt brn, frm - mod frm, sft lip, bkly - sbfssa, stky, wash lip, sil calc. Note : Ahmadi formation is an oil reservoir in some fields. Estimated Fm. PP: 64 - 66 pcf (1.03 - 1.06 gm/cc) EMW Recommended drilling MW: 72 - 74 pcf (1.15 - 1.19 gm/cc)
						Maududd Member : It is composed mainly of Limestone occasionally with very thin stringers of Shale. It is picked by appearance of Limestone occasionally with Increase of Back Ground Gas with Fast ROP. Average TVD Thickness of Maududd Formation : +/- 110 m. Unconformity Surface is available at Base of Maududd formation (Ahmadi Fm.) which is leads of Missing Wara Formation & go directly from Ahmadi formation to Maududd formation at South IRAQ. Limestone : Tansh wh, off wh, crmy wh, occ/ lt brn, mod hd, cryptoxln - v.fxn xln, w/ free pyr, gluc lip, vuggy lip, occ/ sndy, sil dolc, sil arg lip, no vis por under visual exam. Shale : Dk grnsh gy, brnsh gy, occ/ lt - dk gy, frm - mod frm, flky - sb flky, occ/ sb bkly, pyrc lip, mod - hi calc. Note : Maududd formation is an oil reservoir in some fields. Estimated Fm. PP: 64 - 67 pcf (1.03 - 1.07 gm/cc) EMW Recommended drilling MW: 72 - 74 pcf (1.15 - 1.19 gm/cc)
	CRETACEOUS	LOWER	ALBIAN			Nahr Umr Formation : It is composed mainly of Sandstone interbedded with streaks of (Shale, Siltstone & Limestone). It is picked by appearance of Sandstone and/or Shale occasionally with Increase of Back Ground Gas with Fast ROP. Average TVD Thickness of Nahr Umr Formation : +/- 260 m. Caving Shale may occur while drilling Nahr Umr Formation. Sandstone : Lse qtz gr, transl - transp, crl, lt gy, occ/ lt brn, mod hd, fribl - mod hd, fn - vfn gr, occ/ med gr, sb ang - pr - mod strid, sil calc cmt, occ/ w/ arg mtr, w/ free pyr, gluc lip, occ/ w/ TR of Coal, no vis por under visual exam. Shale : Dk grnsh gy, brnsh gy, frm - mod frm, bkly - sb flky, sil arg, gluc lip, non - sil calc, stly lip occ/ grgd to Siltst. Limestone : Lt gy - med gy, tansh wh - lt tan, occ/ lt brn, mod hd, bkly - sb bkly, microxln - v.fxn xln, mod - hi arg, no vis por under visual exam. Siltstone : Lt - med gy, brnsh gy, occ/ lt brn, mod hd, bkly - sb bkly, gluc lip, rr pyrc, non calc, no vis por under visual exam. Note : Nahr Umr formation is an oil reservoir in some fields. Estimated Fm. PP: 64 - 67 pcf (1.03 - 1.07 gm/cc) EMW Recommended drilling MW: 73 - 74 pcf (1.17 - 1.19 gm/cc)
						Shuaiba Formation : It is composed mainly of Carbonates (Limestone and/or Dolomite). It is picked by appearance of Limestone and/or Dolomite with Fast ROP. Average TVD Thickness of Shuaiba Formation : +/- 85 m. partial / Complete Mud losses of circulation is highly possible while drilling Shuaiba Formation. Limestone : wh, off wh, chky, occ/ crmy wh, mod hd, microxln - v.fxn xln, occ/ vuggy, hi dolc occ/ grgd to Dolomitic Limestone, w/ tr of free pyr, no vis por under visual exam. Dolomite : Tan, brn, lt brn, off wh, mod hd, v.fxn xln - mlc xln, suc tex lip, occ/ vuggy, w/ tr of free pyr, no vis por under visual exam. Note : Shuaiba formation is an oil bearing reservoir in some fields. It is highly recommended to control on ROP (Minimize ROP) while drilling Shuaiba formation as it is a highly porous formation to avoid formation break down. Estimated Fm. PP: 66 - 67 pcf (1.06 - 1.07 gm/cc) EMW Recommended drilling MW: 73 - 74 pcf (1.17 - 1.19 gm/cc)
						Zubair Formation : It is composed mainly of (Sandstone & Shale) intercalation with Minor streaks of (Limestone & Siltstone). It is picked by appearance of Sandstone and/or Shale with Slow ROP. Average TVD Thickness of Zubair Formation : +/- 425 m. Caving Shale may occur while drilling Zubair Formation. Sandstone : Lse qtz gr, clse, transl - transp, wh, off wh, occ/ lt brn, mod hd, fribl lip, fn - vfn gr, occ/ med - crs gr, sb ang - pr - mod strid, poorly - mod strid, sil calc cmt, rr glauc, pyrc lip, no vis por under visual exam. Siltstone : Lt gy, off wh, lt - dk brn, brnsh gy, sft - mod frm, bkly - sb bkly, rr pyrc, occ/ grad to v fn sst, non - sil calc. Shale : Dk grnsh gy, dk gy, wh, bkly - brnsh gy - brn, sft - mod frm, fssa - sb fssa, stky lip, occ/ w/ free pyr, washable, sil - hi calc. Limestone : Wh - tansh wh, lt brn, occ/ lt gy, mod hd - hd, cryptoxln - microxln, arg, gluc lip, no vis por under visual exam. Zubair Formation Casing Point : It is a regular casing point, It is recommended to set the casing +/- 60 m into Zubair Formation Top to seal off the swelling shale of upper Nahr Umr formation. Note : Sandstone of Zubair formation is an oil reservoir in some fields. Estimated Fm. PP: 66 - 79 pcf (1.06 - 1.27 gm/cc) EMW Recommended drilling MW: 73 - 90 pcf (1.17 - 1.44 gm/cc)
						Ratawi Formation : It is composed mainly of Shale & Limestone interbeds. It is picked by appearance of Shale with Slow ROP. Average TVD Thickness of Ratawi Formation : +/- 262 m. H2S & Caving Shale may occur while drilling Ratawi Formation. Ratawi Formation is divided into 2 parts (upper part is composed mainly of Shale with minor streaks of Limestone & the lower part is composed mainly of Limestone with minor streaks of Shale). Shale : Grnsh gy, dk brnsh gy, dk gy, mod frm - frm, bkly - sb bkly occ/ sb sille, sil calc. Limestone : Wh - off wh, brnsh gy, lt brn, mod hd - hd, cryptoxln - microxln, sil - mod arg, no vis por under visual exam. Note : Limestone section of Ratawi formation is an oil reservoir in some fields. Estimated Fm. PP: 74 - 101 pcf (1.19 - 1.62 gm/cc) EMW Recommended drilling MW: 79 - 106 pcf (1.27 - 1.70 gm/cc)
		UPPER	PORTLANDIAN	Upper Part		Gotnia Formation : It is composed mainly of Anhydrite & Salt intercalation with minor streaks of (Limestone & Shale). It is picked by appearance of Anhydrite with Slow ROP. Average TVD Thickness of Gotnia Formation : +/- 285 m. Upper part of Gotnia Formation: Occasionally called by Hith formation, It is composed mainly of Anhydrite intercalation with minor streaks of (Limestone & Shale). It is picked by appearance of Anhydrite with Slow ROP. Average TVD Thickness of Upper part of Gotnia Formation : +/- 63 m. Anhydrite : Miky wh, wh, occ/ off wh, rr smky wh, mod hd, occ/ sft, amorph, psd, chky tex lip, washable. Limestone : Dk brn - brnsh gy, dk gy, occ/ lt gy, mod hd - hd, cryptoxln - v.fxn xln, hi arg, no vis por under visual exam. Shale : Bk gy - dk brnsh gy, blk, occ/ lt brnsh gy, frm - mod frm, sb flky - sb flky, sil - mod calc, occ/ hi calc grgd to hi arg Lst. Gotnia Formation Casing Point : It is a common casing point as it is highly recommended to set the casing +/- 10 m into Gotnia formation top or prior drilling Salt beds as it's highly recommended to increase Mud Weight prior drilling Salt beds to avoid up normal high pore pressure, it is a consider a critical casing point which is require to not drill more than +/- 10 m into the first Anhydrite bed at its top. Estimated Fm. PP: 97 - 131 pcf (1.55 - 2.10 gm/cc) EMW Recommended drilling MW: 103 - 144 pcf (1.65 - 2.31 gm/cc)
						Lower part of Gotnia Formation: It is composed mainly of Salt & Anhydrite intercalation with minor streaks of (Limestone & Shale). It is picked by appearance of Salt with very Fast ROP. Average TVD Thickness of Lower part of Gotnia Formation : +/- 210 m Lower part of Gotnia Fm. is Lagoon environment that's why it has a variable TVD thickness +/- 51 m up to +/- 427 m). Water flow associated with (Gas & H2S) & Stuck of Drill String due to Salt mobility behavior in case of hydrostatic pressure is less than formation pore pressure may occur while drilling Lower part of Gotnia Formation. Lower part of Gotnia formation is most critical formation while drilling & it's require to drill & set casing ASAP at Top of Najmah Formation to seal off Gotnia formation to avoid it's huge drilling problems. Salt : Clss, tensl - transp, wh, clear, mod hd, xln, sol. Anhydrite : Miky wh, wh, occ/ off wh, rr smky wh, mod hd, occ/ sft, amorph, psd, chky tex lip, washable. Limestone : Dk gy - blk, dk brnsh gy, mod hd - hd, cryptoxln, hi arg occ/ grgd to hi calc Shale, no vis por under visual exam. Shale : Dk gy - dk brnsh gy, blk, occ/ lt brnsh gy, frm - mod frm, sb flky - sb flky, sil - mod calc, occ/ hi calc grgd to hi arg Lst. Note : Gotnia formation has Up normal High Pore pressure & it is contain mainly of 3 Salt beds separated by 3 Anhydrite streaks & impeded by thin layers of (limestone & Shale). It is occasionally contain mainly of 4 Salt beds separated by 4 Anhydrite streaks & impeded by thin layers of (limestone & Shale), ROP is faster while drilling Salt & it became Slow while drilling Anhydrite. Estimated Fm. PP: 120 - 148 pcf (1.52 - 2.37 gm/cc) EMW Recommended drilling MW: 142 - 152 pcf (2.27 - 2.44 gm/cc)
			OXFORDIAN	NAJMAH / NAO KELIKAN		Najmah / Nao kelikan Formation : It is composed mainly of Limestone & Shale interbeds. It is picked by appearance of Limestone with increase of Back Ground Gas with Fast ROP. Najmah Formation is divided into 2 parts (upper part is composed mainly of Limestone & the lower part is composed mainly of Shale with streaks of Limestone). Average TVD Thickness of Najmah Formation : +/- 41 m. Limestone : Lt gy, tansh gy - tan, gysh wh, lt brn, occ/ brnsh gy, mod hd - hd, microxln - v.fxn xln, non - sil arg, sil dolc lip, no vis por under visual exam. Shale : blk, brn, frm - mod frm, bkly - sb bkly, w/ blk carb matter, bituminous, hi calc occ/ grgd to hi arg Lst. Najmah / Nao kelikan Formation Casing Point : It is a regular & Most critical csd point, it is highly recommended to set the casing +/- 5 m into Najmah Formation top to seal off Gotnia formation which has Up normal high pore pressure & also reduce Mud Weight while drilling Najmah formation as the Gotnia formation pressure of Najmah formation is less than the above formation pore pressure of Gotnia formation. Note : Najmah formation casing point is picked by appearance of Limestone with light colors as the Limestone of the above Gotnia formation with Dark colors. Note : Najmah formation is an oil reservoir in some fields. Estimated Fm. PP: 112 - 129 pcf (1.79 - 2.07 gm/cc) EMW Recommended drilling MW: 118 - 133 pcf (1.89 - 2.13 gm/cc)
						Sarjelu Formation : It is composed mainly of Limestone & Shale interbeds. It is picked by appearance of Argillaceous Limestone with Slow ROP. Sarjelu Formation is divided into 2 parts (upper part is composed mainly of highly argillaceous Limestone & the lower part is composed mainly of highly calcareous Shale with streaks of Limestone). Average TVD Thickness of Sarjelu Formation : +/- 62 m. Limestone : Brnsh gy - dk brn, crmy wh, lt gy, occ/ dk gry, mod hd - hd, microxln - cryptoxln, mod - hi arg, no vis por under visual exam. Shale : Gy, lt gy, grnsh gy, occ/ dk grn, frm - mod frm, sb flky - sb flky, stky, washable, pyrc lip, hi calc, grgd to hi arg Lst and/or Marl. Note : Sarjelu formation is an oil reservoir in some fields. Estimated Fm. PP: 114 - 135 pcf (1.83 - 2.16 gm/cc) EMW Recommended drilling MW: 118 - 139 pcf (1.89 - 2.23 gm/cc)
	JURASSIC	MIDDLE	CALLOVIAN	SARJELU		Alan Formation : It is composed mainly of Limestone with minor streaks of Anhydrite. It is picked by appearance of Limestone occasionally with Back Ground Gas with Slow ROP. Average TVD Thickness of Alan Formation : +/- 80 m. Salt water flow may occur while drilling Alan Formation. Limestone : Dk gy, brn - brnsh gy, occ/ lt gy, mod hd - hd, microxln - cryptoxln, mod - hi arg, no vis por under visual exam. Anhydrite : Miky wh, wh, occ/ off wh, rr smky wh, mod hd, occ/ sft, sil calc, amorph, psd, hi washable. Alan Formation Casing Point : It is not a regular casing point, It is recommended to set the casing +/- 50 m (or) +/- 75 m into Alan formation. Estimated Fm. PP: 105 - 135 pcf (1.68 - 2.16 gm/cc) EMW Recommended drilling MW: 108 - 138 pcf (1.73 - 2.21 gm/cc)
						Mus Formation : It is composed mainly of Limestone & occasionally with minor streaks of Anhydrite. It is picked by appearance of Thick bed of Limestone below Anhydrite streaks of the upper formation with Fast ROP. Average TVD Thickness of Mus Formation : +/- 110 m. H2S may occur while drilling Mus Formation. Limestone : Lt gy, lt brn - lt brnsh gy, mod hd - hd, microxln - v.fxn xln, sil arg, no vis por under visual exam. Anhydrite : Miky wh, wh, occ/ off wh, rr smky wh, mod hd, occ/ sft, sil calc, amorph, psd, hi washable. Note : Mus formation is an oil reservoir in some fields. Estimated Fm. PP: 105 - 137 pcf (1.68 - 2.19 gm/cc) EMW Recommended drilling MW: 108 - 138 pcf (1.73 - 2.21 gm/cc)
						Adaiyah Formation : It is composed mainly of Anhydrite with streaks of Limestone. It is picked by appearance of Thick bed of Anhydrite with Slow ROP. Average TVD Thickness of Adaiyah Formation : +/- 130 m. H2S may occur while drilling Adaiyah Formation. Anhydrite : Miky wh, wh, occ/ off wh, rr smky wh, mod hd, occ/ sft, sil calc, amorph, psd, hi washable. Limestone : Lt gy, lt brn - lt brnsh gy, mod hd - hd, microxln - v.fxn xln, sil arg, no vis por under visual exam. Estimated Fm. PP: 105 - 137 pcf (1.68 - 2.19 gm/cc) EMW Recommended drilling MW: 108 - 138 pcf (1.73 - 2